Enefit: Predict Energy Behavior of Prosumers

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1 Introduction

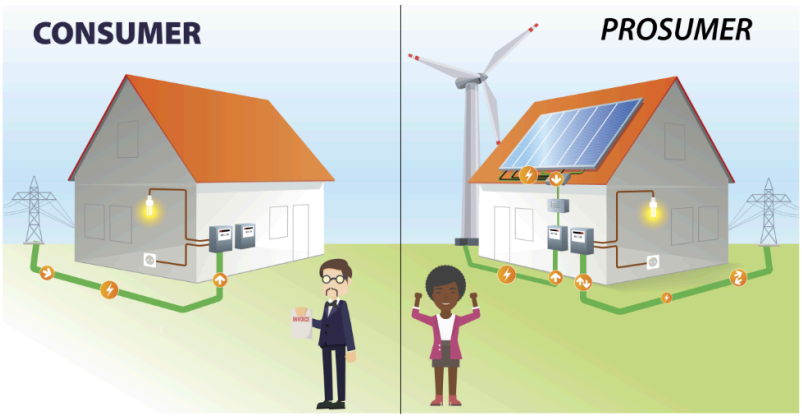


Figure 1: 3-D image that shows the difference between a consumer and a prosumer. Image from https://www.energy.gov/eere/articles/consumer-vs-prosumer-whats-difference

Energy prosumers are individuals, businesses, or organizations that both consume and produce energy. This concept represents a shift from the traditional model where consumers simply purchase energy from utilities and rely on centralized power generation sources. Energy prosumers are actively involved in the energy ecosystem by generating their own electricity, typically through renewable energy sources like solar panels or wind turbines etc.. They also consume energy from the grid when their own generation is insufficient to meet their needs. The number of prosumers is rapidly increasing, associated with higher energy imbalance - increased operational costs, potential grid instability, and inefficient use of energy resources. If not taken into account, the increasing energy imbalance caused by prosumers can cause logistical and financial problems for energy companies.

2 Background

The goal of this project is to create an energy prediction model of prosumers to reduce energy imbalance costs. If solved, it would reduce the imbalance costs, improve the reliability of the grid, and make the integration of prosumers into the energy system more efficient and sustainable. The model accuracy would be determined using Mean Absolute Error (MAE).

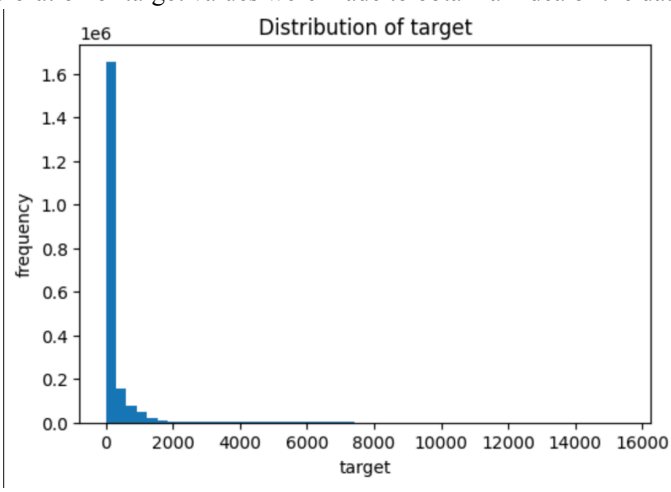
$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - x_i\right|$$

Where:
n is the total number of data points
y_i is the predicted value for the data point i
x_i is the observed value for data point i

The dataset is derived from Estonian energy customers who have installed solar panels. It gives us access to weather data, relevant energy prices and records of installed photovoltaic capacity alongside records of target, the consumption or production amount of a customer and the value we want to predict. Some of the core data files include information regarding historical production and consumption of energy across counties, by both individual and business prosumers. For instance, train.csv contains features for target energy amount, datetime for meeting/setting that expectation, and whether that energy is consumed or produced, as well as unique identifiers for individuals called prediction_unit_id. Some other .csv files include gas prices, electricity prices, details about the client and how their solar panel setups look like. There also exists .csv files for historical and regular weather forecast data which is key for solar energy production. The dataset is heavily reliant on the datetime as that ties many of the important features together. The datetime could be 'unpacked' to introduce newer features like day, year, month etc. and utilize those to identify potential temporal trends. Furthermore, since this competition is a unique blend of time-series forecasting and model prediction, more recent data could play a big part in deciding future values (for example, energy prices may be less suspect to change from week to week as opposed to month-to-month, influencing target values). Hence, lagging the data was important in modeling and it also helped create new features that our model can use. Once a clean dataset with feature engineering was obtained, it was split into three components- train, test and validation (see notebook for more details).

3 Exploratory Data Analysis

Initial exploration of target values were made to obtain an idea of the data that needed to be analyzed.



statistic	value
---	---
str	f64
count	2.017824e6
null_count	528.0
mean	274.85556
std	909.502378
min	0.0
25%	0.378
50%	31.133
75%	180.206
max	15480.274

Figure 2: Histogram of target distribution and statistics

As expected, a majority of the target values are less than 1000-500 with a mean of ~274, indicating the existence of small-time businesses, individual entities and rarely any large scale operational entities that require high energy. After this, a visual representation of the daily average of the target variable over time was plotted to spot any potential temporal trends.

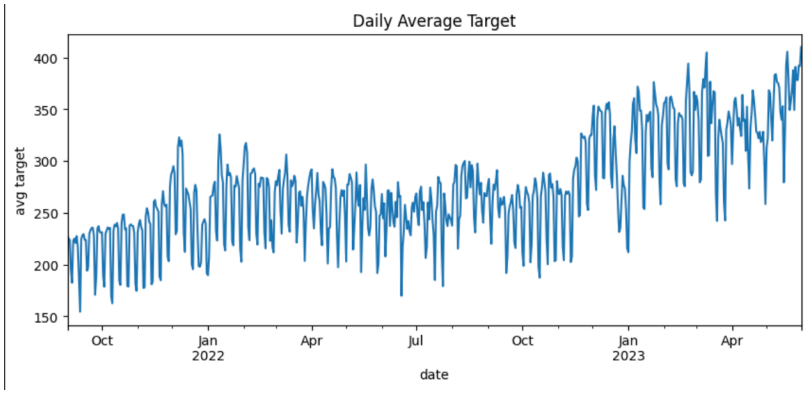


Figure 3: Plot of avg. target vs time

Overall, target values seem to be increasing as time goes by suggesting a nationwide increase in energy supply and demand. The recurring up and down pattern of the graph suggests a strong weekly seasonality (maybe having a higher target value during weekends vs weekdays) and confirms the suspicion of underlying temporal trends noted earlier.

In order to explore these temporal patterns, Fast Fourier Transform (FFT) was applied on multiple segments, where a segment of the form 'a_b_c_d' represents a categorical combination of the following features: county, is_business, product_type and is_consumtion respectively. FFT is incredibly helpful in breaking down information from the time domain into a frequency domain to identify how often patterns repeat. This can help reveal cycles through a spike in frequency for that particular time period.

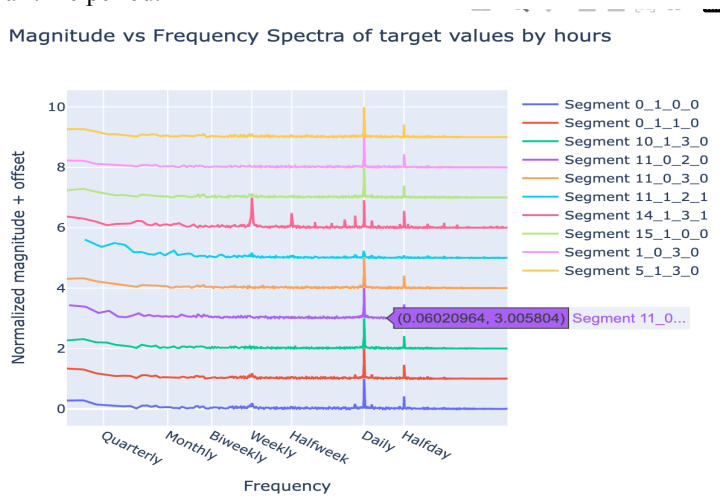


Figure 4: Magnitude vs Frequency spectra of target values, where the periods were analyzed in hours to better identify smaller changes. Y-axis shows normalized, positive target magnitudes while X-axis shows labeled tick marks for time periods

It can be seen from Figure 4 that there are almost always peaks for daily and half-day magnitudes, with some peaks for weekly magnitudes yet again indicating strong short term influence on target value spikes. Ideally , the next and final set of data analysis lies in finding out which features contribute to the spikes observed in Figure 4. Using the same FFT technique, a few features were hand selected intuitively and observed to see any patterns.

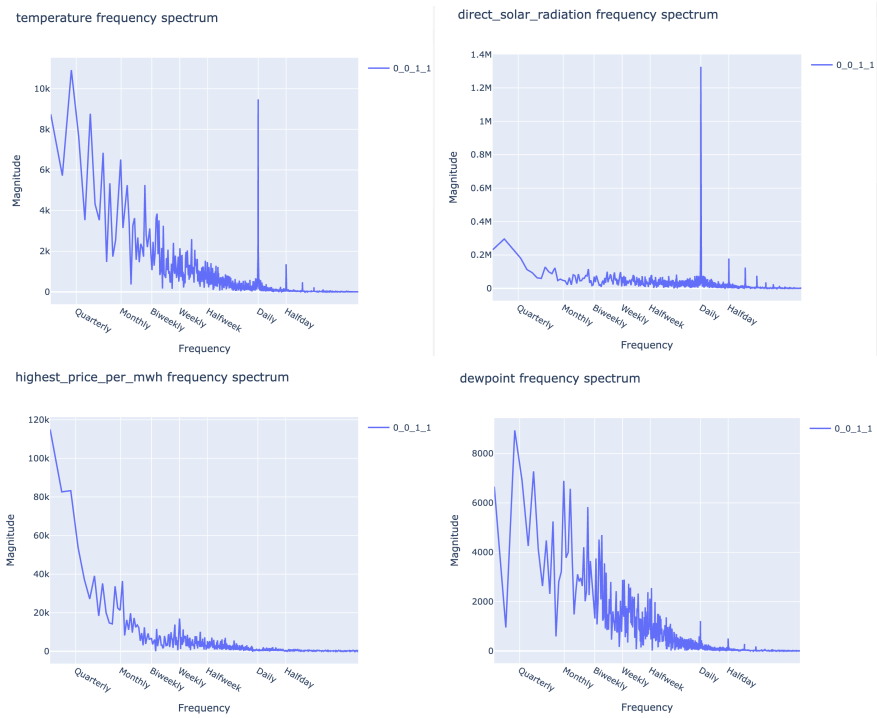


Figure 5: Magnitude vs Frequency spectrum for different features for the same segment

While temperature and direct solar radiation show prominent spikes in the daily time spectrum, indicating their strong presence, some other features are shown to not have as prominent of an impact, positive or adverse. However, with temperature and direct solar radiation essentially meaning the same fundamental quality, this led to the question: would dimensionality reduction and other novel methods help assist us with data modeling?

4 Approach

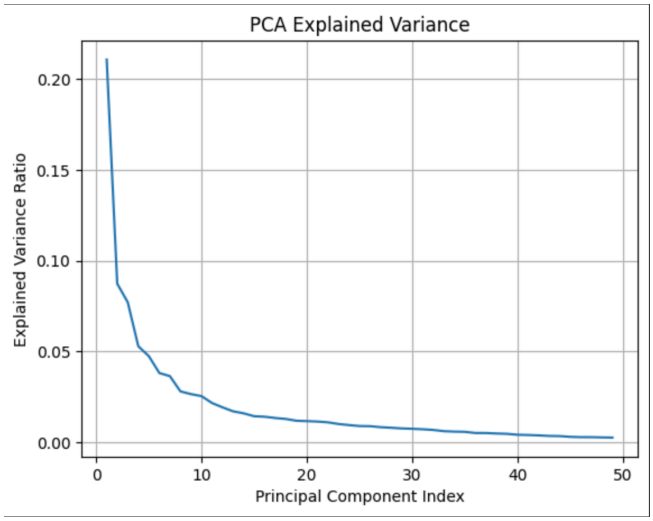
4.1 Lagged and forecasted data

During feature generation, it was important to carefully think about what type of data would realistically be available and what type of features, if lagged appropriately, can benefit prosumers. Features like gas prices, electricity prices, weather forecasting, are subject to quick change and would benefit a model from forecasting. Features like target values, if lagged to maybe 2 days or 7 days ago, can help give a model a stronger 'remembering power' to respond to the weekly and daily temporal changes we noticed in our data analysis. Lagging and forecasting, as a result, is carefully introduced to select features to ensure that the model can successfully prioritize data in a more recent time frame since our target values fluctuate on a weekly basis while still drawing information from the entire timeline of data available. This means that data at the very beginning may not have prior data to work with. As a result, the dataset was imputed using mean values to account for NaNs that can prevent models from predicting target values. Once feature engineering was done, the dataset had a sum value of ~140 features and ~2 million rows worth of data to work with.

4.2 Principal Component Analysis

For post-processing, and after observing some feature similarities, the first approach I tried was to reduce dimensions using Principal Component Analysis (PCA). However, before applying PCA, some features which were categorical like segment and time strings like datetime needed to be excluded to prevent PCA confusion. This helped reduce dimensionality down by ~64% while keeping 95% of the original data variance. The new shape of the

dataset had 50 dimensions after adding back categorical features.




```
Top 15 features contributing to PC1:
temperature_hl_2d                          0.160953
temperature_fl                             0.160856
temperature_hd_2d                          0.159743
temperature                                0.157639
cos(dayofyear)                             0.156995
surface_solar_radiation_downwards_fd_7d    0.156085
temperature_fl_7d                          0.155305
shortwave_radiation_hd_7d                  0.155202
temperature_hl_7d                          0.155024
surface_solar_radiation_downwards_fl_7d    0.153861
temperature_hd_7d                          0.153291
shortwave_radiation_hl_7d                  0.152929
temperature_fd_7d                          0.150657
direct_solar_radiation_fd_7d               0.150264
direct_solar_radiation_hd_7d               0.150220
```

Figure 6: (Left) Variance explained for principal components, follows an elbow curve where it captures a majority of the variance in the first 10 principal components and then falls off.
(Right) List of features that contributed to the first principal component and its resultant variance

However, after comparing PCA dataset with the raw dataset on some baseline models (DummyRegressor with strategy='mean', LinearRegressor and HistGradientBoosting), the raw dataset performed much better on the baseline models than the PCA dataset. Keeping approach 4.1 in mind, I also tried only including more relevant data by excluding data points before 2022. However, that only harmed the MAE score and hence the final approach was done based on the raw dataset with all the data points.

4.3 LightGBM and XGBRegressor

With the existence of hundreds of features that generate a complex tabular data, and PCA giving adverse results on baseline models than the original dataset, it was imperative to choose advanced models that can capture robust relationships between features, for example: "when it's forecasted to be cold on the weekend and my past weekend target value was high, chances are that future target usage will be high as well". LightGBM and XGBRegressor are two advanced tree-based models that can efficiently capture advanced non-linear relationships well. This, combined with the fact that lagged/forecasted values are feature engineered in the dataset, also makes these models time-series friendly and much more powerful to pick up temporal trends.
XGBRegressor was trained first on the feature engineered dataset without any particular hyperparameter tuning followed by the LightGBM model, which had a hyperparameter tuning phase through Optuna followed by a training phase with those hyperparameters. The feature importances after training are as follows:

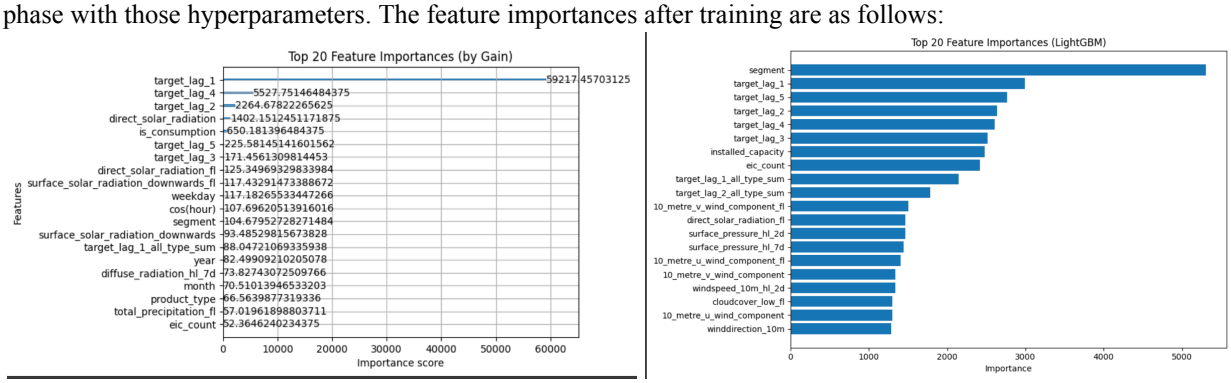


Figure 7: Feature importances of XGBRegressor and LightGBMRegressor models after training

4.4 Resulting score

The final score was determined by a non-PCA, feature engineered dataset trained on LightGBM and XGBRegressor models. The XGBRegressor model achieved an MAE score of 33.4159 on the test set while the LightGBM model which was hyperparameter tuned achieved a score of 23.1683. This shows the importance of hyperparameter tuning for even advanced models.

Note that this is based on the split performed on the existing dataset and not on hidden test cases/data used for submission on the competition page, because I had trouble importing the API that updates testcases onto my workspace. It can be assumed that the model will perform worse than the current scores on newer unseen data.

5 Ablation study

I attempted to model the data without LightGBM and XGBRegressor and replaced them with dummy/baseline models and also tried using the dummy models on the PCA dataset vs the raw features. The results are as follows:

```
DummyMean (Raw Features):
  Val MAE = 370.8971, R² = -0.0000
LinearRegression (Raw Features):
  Val MAE = 69.2588, R² = 0.9380
HistGradientBoosting (Raw Features):
  Val MAE = 41.7458, R² = 0.9800
```

```
DummyMean: MAE = 370.8971, R² = -0.0000
LinearRegression: MAE = 86.0199, R² = 0.9173
HistGradientBoosting: MAE = 81.4068, R² = 0.9517
```

Figure 8: (Top) Dummy/Baseline models on original dataset
(Bottom) Dummy/Baseline models on PCA dataset
Both were on the validation set and not the training dataset to conserve memory usage.

6 Future Directions

This competition was a unique blend of time-series data and feature lagging/engineering that I have never done before, so implementing this was definitely a challenge and would not be possible without the help of numerous discussion posts on the Kaggle community. I definitely feel like I have some possible data leaks due to my inexperience with dealing with the conversion of time series data and lagged feature creation. Something to try in the future would most definitely be a better feature engineering method that can address my foundational shortcomings after getting more acquainted with time-series data. I think it would be interesting to do ensemble learning and see what my result is. It would also be interesting to see if Neural Networks may impact the MAE value.

References

[1] https://www.kaggle.com/competitions/predict-energy-behavior-of-prosumers/discussion/472537

[2] https://www.energy.gov/eere/articles/consumer-vs-prosumer-whats-difference

[3] https://lightgbm.readthedocs.io/en/v3.2.1/pythonapi/lightgbm.LGBMRegressor.html

[4] https://www.kaggle.com/competitions/predict-energy-behavior-of-prosumers/discussion/472793

[5] https://www.kaggle.com/competitions/predict-energy-behavior-of-prosumers/discussion/499938